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Project Title:	Investigation into the Quantification and Control of Steady and Unsteady Flow Phenomena in Passive Gas-Liquid Cyclonic Separators		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
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Space Biology Special Category:	None		
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Zip Code:	44106	**Congressional District:**	11
Comments:			
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No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	McQuillen, John	**Contact Phone:**	216-433-2876
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Flight Program:			
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Key Personnel Changes/Previous PI:			
COI Name (Institution):	Kadambi, Jaikrishnan (Case Western Reserve University)		
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	Phase separators are devices designed to divide flows of intermingling gas and liquid into separate streams that are individually purely liquid or purely gas. Separation of the respective phases is of considerable interest to NASA as many spacecraft systems can only function effectively if the flow they receive is composed exclusively of the one phase that they are designed to accommodate. While separation occurs naturally on the earth, the absence of buoyancy in a microgravity environment allows the undesired mixtures of gas and liquid to remain combined unopposed. Due to this reluctance of the phases to separate, installation of some type of separation device becomes supremely important for many of the fluid systems that are used by NASA. Cyclonic separators provide the gas-liquid separatory action by swirling the multiphase flow – causing the gas to accumulate along the axis of the vortex as the denser liquid is forced to the walls – thereby allowing segregated extraction of the respective phases. Passive cyclonic separators utilize only the		

Task Description:	inertia of the incoming flow to accomplish this task and in doing so can avoid the moving parts and reliability concerns of their active brethren. While devices of this type have been successfully employed in experimental settings, more complete investigations and analyses are required before these devices can ever be seriously considered for use in operational space hardware. The objective of this study is to facilitate the maturation of passive cyclonic separation technology through exhaustive research of a newly designed gas-liquid phase separator evolved from NASA Glenn Research Center's (GRC) existing Cascade Cyclonic Separation Device (CSD-C). Combined experimental and numerical analyses will seek to quantitatively assess and delimit the range of operability of the new separator including the transient and steady-state instability behavior of the device with respect to both device and system-level conduct – features that have yet to be completely explored in any such device. The approach is first to assess the range of operability of the new separator design through the careful construction of detailed flow separation maps using data gathered from reduced-gravity flight experimentation in conjunction with ancillary computational studies. Second, the transient and stability behavior of the device will be observed and analyzed. Special attention will be paid both to device-level conduct and also to behaviors that could manifest dysfunction elsewhere at the complete system level. A novel extension of the phase-field computational multiphase fluid dynamics (CMFD) method has been coupled to 1-D system-level multiphase solvers in order to permit discernment of these flow features. Lastly, with the ensuing understanding of the causes of any performance-affecting responses, new ways to control transient and unstable operations will be sought; in particular, the insight acquired should facilitate further new and original design modifications to improve the performance of microgravity phase separation devices. Validation and verification of the numerical results are expected to permit the use of CMFD as a reliable design tool to further assist in the creation and analysis of new designs. The combination of the research activity included in this grant with the complementary activity of Dynaflow Inc. of Jessup, MD (Grant NNX09AI35G) has evolved into the Two-Phase Flow Separator Experiment (TPFSE), a project that has been elevated to flight status. TPFSE represents a comprehensive investigation of passive cyclone separation technology – an investigation made possible by capabilities that only NASA and the International Space Station can offer.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Phase separation has been identified as a critical mission-enabling capability for spacecraft systems ranging from power conversion to Environmental Control and Life Support Systems. As these systems have little tolerance for failure, extensive investigations and analyses are required before these devices can ever be seriously considered for employment in operational space hardware. Due to the advantages associated with passive vortex devices, cyclonic separators are also finding increasing use in terrestrial applications, such as in the petroleum industry where separators of this type are considered to be an emerging technology. The detailed study of the features of the complicated vortex-driven flow along with the modeling techniques under development for the microgravity investigations could equally be exploited for the advancement of these more earthly applications.
Task Progress:	Separator Design The project plan initially concerned experimentation and analysis with respect to the existing GRC CSD-C separator, but a design evolution, the Case Western Reserve University Separator (hereafter referred to as CWRU-SEP), that is expected to perform better for the experimental and operational needs of the project has been developed instead. The design improvements of CWRU-SEP arose by capitalizing on the results of our previous investigations into passive-separation technology (funded via the NASA Graduate Student Research Program [GSRP]). Numerical studies completed both during the first year of this grant and during the GSRP period of funding have permitted optimizations to the device interior, the inlet design, and the outlet design of the device. Strengthening of the vortex structure within the separator housing was one of the main design parameters therein. Additional management of the pressure balance between the outlets is achieved by the inclusion of a swirl arrestor at the liquid outlet in order to partially curtail the free vortex beneath the baffle plate. The new design is also intended to be more modular than the previous CSD-C. Interchangeable swirl arrestor plates and injection plates have been included to permit the easy addition of further design enhancements that are currently under investigation. Various combinations of geometric parameters regarding the precise form of the conical body are also possible. The manners in which these alterations affect observability of the flow within the device have also been taken into consideration. The finalized CWRU-SEP design was fabricated and has been operated since early in the second year of this grant. Numerical Modeling The CMFD studies of the fluid physics within the separation device conducted during the first year of the grant have been continued. The CWRU-SEP design optimization studies utilizing multifluid treatments have continued, but more advanced techniques have been created to improve predictive capabilities. In particular, a custom phase-field surface implementation has been developed and has been used to simulate some of the capillary instabilities that affect the functionality of the separator – features that could not be captured with the previous modeling efforts. System-level conduct has been found to be crucial to the proper functionality of the separator – as such, efforts have been made to model the upstream and downstream components that are invariably present. To do so, one-dimensional two-phase flow modeling techniques have been coupled to the device-level three-dimensional simulations. Compressible HEM and multifluid approaches have both been employed. Two-phase Flow Experimental Apparatuses Operations of the combination open/closed two-phase flow loop (constructed at CWRU during the first year) have continued during the second year. This work is has provided guidance regarding subsequent 0-g operation. The loop has been permitted the use of a deconstructivist approach with respect to the separator design. For example, investigations into the individual components, such as the injection nozzles, have also been performed. Experimentation with simplified straight-cylinder vortex separators and with the CWRU-SEP design has been performed

	using this loop to understand the fluid physics and to permit validation and verification of CFD approaches under development. Measurements of axial velocity and the tangential velocity distributions in the liquid layer were carried out using ultrasound techniques (pulsed Doppler ultrasonic velocimetry). Good agreement between experimental and computational results has been observed. Computational results have also reproduced system-level instability behavior and have matched pressure oscillations that were arising due to representative density-wave oscillations. Work on refurbishing the Two-Phase Flow Experiment (TPFE) apparatus at GRC has been begun in order to facilitate more sophisticated instrumentation work and the progression to microgravity flights. A flight campaign (using the TPFE rig as the basis) is tentatively scheduled for next February. TPFSE As this research activity has evolved into the Two-Phase Flow Separator Experiment (TPFSE), work has continued in accordance with the elevation of the project to flight status. Definition of the complete experimental requirements has been continued with respect to the objective of the emplacement of the experimental apparatus aboard the International Space Station. The changes to the separator design that necessarily accompany ISS installation are being examined. Collaboration with NASA and contract personnel has furthermore continued in earnest to start defining the basic make-up of the complete system that is to be eventually employed on Station.
Bibliography Type:	Description: (Last Updated: 03/06/2024)
Abstracts for Journals and Proceedings	Hoyt N, Kang M, Kamotani Y, Kadambi J. "Cyclonic Two-Phase Flow Separator Experimentation and Simulation for Use in a Microgravity Environment." To be presented at the 4th International Symposium on Physical Sciences in Space (ISPS4), Bonn, Germany, July 11-15, 2011. 4th International Symposium on Physical Sciences in Space (ISPS4), Bonn, Germany, July 11-15, 2011. Journal of Physics Conference Series, expected publication in last quarter of 2011. , Jul-2011